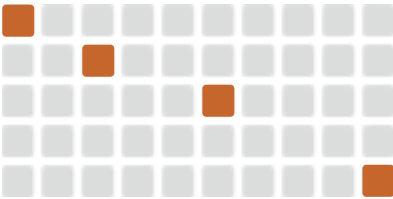


# Assay Chain Architecture

How we build a laboratory information system that follows a gold sample from the drill hole to the certified purity of a doré or bullion bar — and stays sellable to the next lab.

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|          |                               |
|----------|-------------------------------|
| Client   | in-house mine assay lab       |
| Basis    | site walkthrough, August 2026 |
| Status   | draft for internal review     |
| Document | technical foundation, v1.2    |



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## Where this stands

This is the technical foundation, written after the site walkthrough and before the PRD. It records the decisions that are expensive to reverse later, and it is deliberately opinionated so that the team has something concrete to argue with.

Several inputs are still outstanding — instrument makes and models, sample throughput, the acQuire configuration on the geology side, and the current report formats. None of them change the shape below. They change sizing, effort, and sequencing, which is why the release plan in section 12 is expressed as scope rather than dates.

One thing worth stating plainly at the top: the lab records data today in Excel, on paper, and in operators' heads. The value of this system is not that it stores the same numbers in a database. It is that *the number arrives from the instrument that produced it, attached to the sample it came from, with the evidence still attached*. Every decision below serves that.

## The constraint that shapes everything

The brief has two customers: this lab, and every lab we sell to afterwards. That second customer is the one that determines the architecture, because it forbids the shortcut that would otherwise be tempting — hard-coding this lab's batch size, its QC limits, its ID format, its report layout and its workflow into the application.

Everything the client described as “should be configurable” is really one requirement: **the differences between laboratories live in data, not in code**. A new lab is a new configuration set, not a new branch.

#### DECISION D1

**Configuration is versioned data, and every result binds to the version that produced it.**

Methods, workflow stages, batch templates, QC rule sets, ID schemes, report templates, roles and rounding conventions are all stored as configuration objects with version numbers. A QC evaluation stores the identifier of the rule-set version that evaluated it.

This matters more than it sounds. Without it, an authorised user widening a CRM tolerance today silently changes whether a batch from last year passed — which is precisely the kind of retroactive change an ISO/IEC 17025 assessor is looking for. Configuration changes are themselves auditable events.

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**Rejected:** settings tables edited in place. Cheap to build, impossible to defend in an audit, and it makes historical data unreproducible.

## 02

### What the system actually models

Most laboratory systems are built on a single idea: a sample arrives, is tested, produces a result. That model is wrong here, and it is the reason generic LIMS products fit assay labs badly.

A gold sample does not stay one object. Three kilograms of drill core becomes a dried sample, a crushed sample, a split, a pulp, a fifty-gram charge, a lead button, a prill weighed in micrograms, and finally a grade in grams per tonne. Mass leaves at every step, and the reject of each step is retained. If the data model cannot express “this number came from that piece of that sample,” then the traceability requirement cannot be met no matter how good the audit log is.

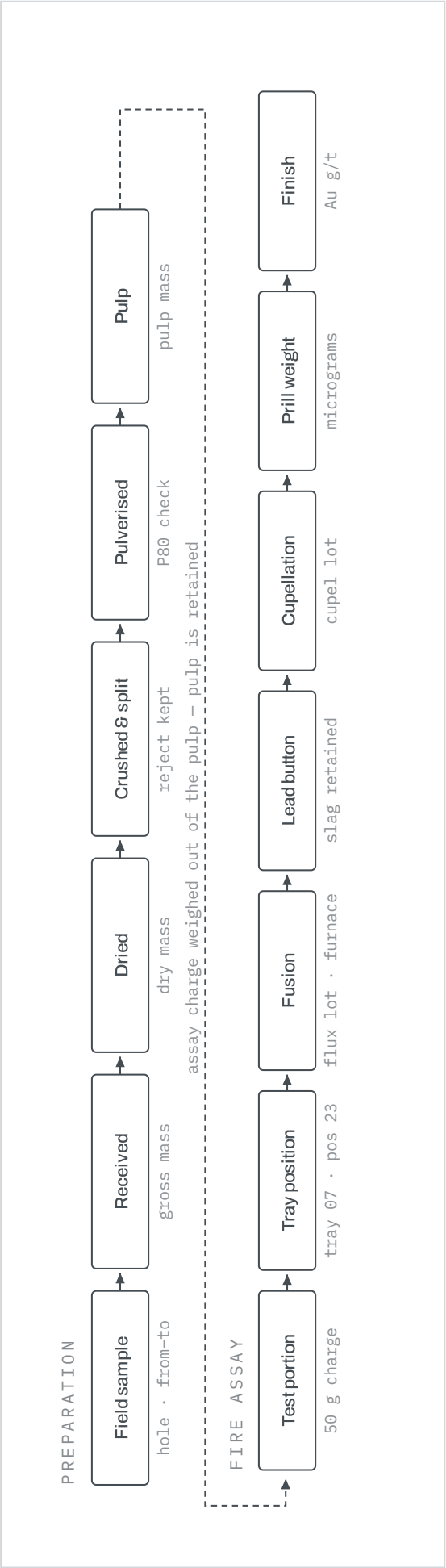
#### DECISION D2

**A sample is a graph of transformations, not a row.**

Every derivation is an edge carrying the operation, the mass in, the mass out and the operator. The graph allows merges as well as splits, because composites and bulk metallurgical samples combine parents.

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**Rejected:** a flat sample table with a parent identifier. It cannot represent composites, and it loses mass balance, which is the only mechanical check we have on whether prep was done correctly.



**FIGURE 1 – The assay chain.** Each box is an object the system stores; each arrow is a derivation edge carrying mass in, mass out, operator and timestamp. The dashed return is the only place the chain narrows dramatically – kilograms of pulp become a fifty-gram charge – which is why the pulp and every reject are retained for re-assay.

## Core entities

|                                                                                                    |                                                                                                    |
|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Dispatch</b><br>A field submission. The unit of transfer between geology and the lab.           | <b>FieldSample</b><br>Hole, from-to depth, type, collector, requested work.                        |
| <b>Sample</b><br>The lab-side object, created on receipt, matched to a FieldSample or provisional. | <b>Derivation</b><br>Parent to child edge. Operation, mass in, mass out, operator.                 |
| <b>TestPortion</b><br>The charge weighed out for a determination.                                  | <b>Batch · Position</b><br>A tray with addressable positions holding portions or QC inserts.       |
| <b>Run</b><br>A fusion, cupellation, weighing, AAS or LECO execution.                              | <b>Measurement</b><br>One captured value with instrument, method, operator, raw payload reference. |
| <b>Result</b><br>A derived, reportable value. Versioned; corrections supersede.                    | <b>QCEvaluation</b><br>Outcome of a rule set against a control, bound to the rule version.         |
| <b>ConsumableLot</b><br>CRM, flux, crucible and cupel lots with certificates.                      | <b>Melt · Bar</b><br>Doré pour, bar serial, gross mass, bar sample, fineness.                      |

Note that `ConsumableLot` is present in version one even though crucible tracking is deferred. The reason is CRMs: a QC evaluation cannot be performed without the certified value and standard deviation from the specific lot certificate. Lot handling has to exist from day one; what is deferred is extending it to crucibles and cupels.

03

## Trays, positions and quality control

The lab batches in fifties. A batch of fifty is almost never fifty client samples — it is samples interleaved with blanks, certified reference materials and duplicates at defined positions. Position is not cosmetic. Furnaces are not uniform, and position-correlated bias is a real diagnostic signal. If the batch is stored as an unordered list, that signal is gone.

#### DECISION D3

### **A batch is a tray with addressable positions, and the QC insert map is configuration.**

The batch template defines the tray size, the position layout, which positions carry which class of control, and the insertion rate. An authorised user edits the template; the change is versioned, and batches already fired keep the template version they were built under.

The QC rule set is likewise configuration: CRM tolerance expressed in standard deviations against the lot certificate, blank threshold, duplicate relative percentage difference limit, insertion frequency, and the action taken when a rule fails. Failure actions must include hold, warn, and quarantine the batch, because different labs draw that line differently.

Approval sits with the manager, as described on the visit. Two rules make that meaningful rather than ceremonial: the person who produced a result cannot be the person who approves it, and a re-assay is never a silent replacement — the original stays, the repeat is linked to it, and both carry the re-assay flag through into reports.

### **Blind quality control**

Geology inserts its own controls into a dispatch, and whether the lab can see them is a setting rather than a fixed policy. That is the right call — but a toggle only means something if it is governed, so the design has to settle three things a plain switch leaves open: who may set it, when it lifts, and what actually enforces it.

#### DECISION D11

**Control visibility has three modes, is set by the party submitting the samples, and can never be changed from the lab side.**

**Visible** — the lab sees that the sample is a control and which lot it is. This is the mode for the lab's own tray inserts.

**Blind** — the lab sees an ordinary sample. Identity reveals automatically on a configured event, normally batch approval. Blindness has an expiry, and the expiry is an event rather than somebody's decision.

**Retained** — never revealed to the lab at all. Only the submitting party and the auditor role ever see the evaluation.

Precedence runs insert over dispatch over tenant default, so geology can set a house policy and still override it for one submission. A tenant policy can also lock the setting so the lab can never be granted visibility — which is what a commercial laboratory's contract with an external client will eventually require.

---

**Rejected:** a visibility flag the lab can toggle for itself. If blindness can be lifted by the party being tested, it is decoration.

#### Blindness is enforced below the interface

If visibility is a screen filter it will leak — through an export, a search result, a report, a validation message or a log line. A blind control returned to a lab-scoped query has to come back as an ordinary sample with the control attributes *absent*, not present and hidden. That rule belongs in the data access layer, where every query path inherits it, and not in the views.

The channels that leak in practice, each of which has to be closed deliberately:

- **Identifiers.** Blind controls draw from the same number sequence as real samples and sit at plausible points in it. A control numbered at the end of the range announces itself.
- **Counts.** A dispatch of fifty shows the lab fifty samples, not forty-eight and two controls.
- **Requested work.** A control asking for a different determination suite than everything around it is a tell. Blind controls carry the same suite as their neighbours.
- **Placement.** The field app scatters inserts around a target rate rather than at an exact interval. A control at every twentieth position is a pattern the bench notices inside a week.
- **Evaluation feedback.** The important one. A failed lab control holds the batch — but a failed *blind* control cannot, because the hold itself reveals it. Blind evaluations route to the submitter, and any re-assay that follows reaches the lab as an ordinary re-assay request carrying no reason that gives it away.

- **The audit trail.** The log necessarily records that a blind control exists. Auditor and geology roles see those entries; lab roles do not. Audit queries are role-scoped like everything else.

## Two quality-control streams, one engine

It follows that there are two independent QC streams sharing the same rule engine and the same lot certificates, differing only in who owns them and who can see them. The lab's own tray inserts are visible, gate batch approval, and hold work when they fail. The submitter's inserts may be blind, are evaluated on the submitter's side, and never gate lab approval — they drive the conversation that follows instead.

Separating the streams also gives the client something they do not have today: a defensible answer to *whose* control failed, which is the first question anyone asks when a result is disputed.

04

## Getting numbers off the instruments

The instrument PCs are on the site network and IT will grant access, which makes this tractable: a small capture agent runs as a Windows service on each instrument PC and pushes what the instrument produces to the edge server.

The agent must handle four capture modes, because lab instruments have not converged on anything: a **serial listener** on RS-232 or USB-serial for balances, which emit a reading on the print or stable-weight signal; a **watched folder** for AAS and LECO software that exports CSV, TXT or XML; a **print-stream capture** for older instruments that can only print; and **manual entry with double-key verification** as the universal fallback, because there will always be one instrument nobody can reach.

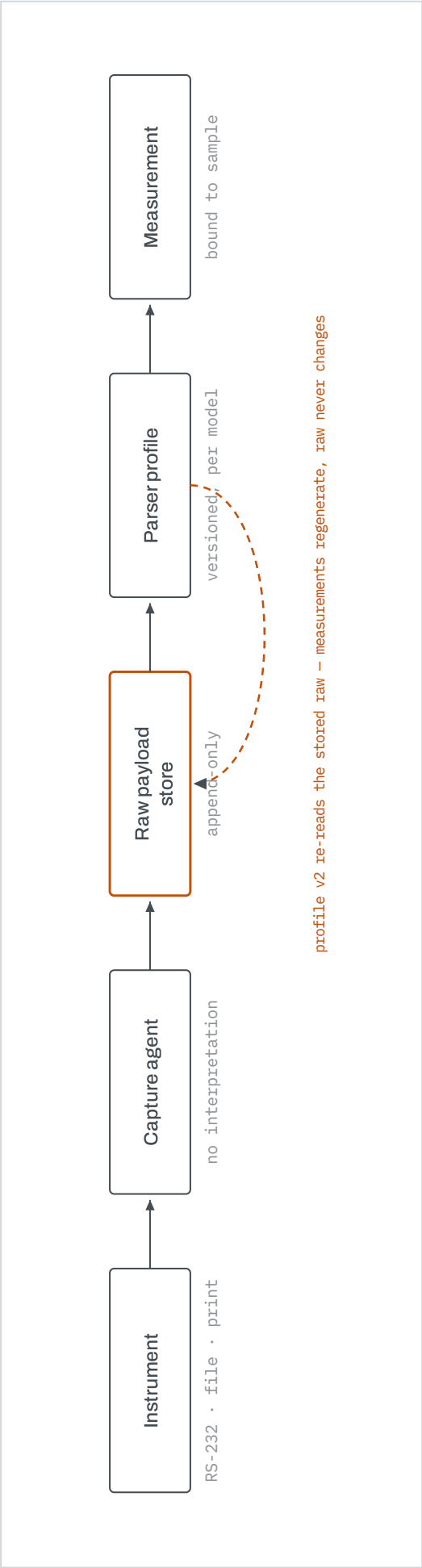
### DECISION D4

**The agent captures raw and interprets nothing. Parsing happens on the server, against a versioned parser profile, and the raw payload is kept forever.**

The agent posts the bytes it received plus instrument identity, timestamp and source. The server parses using a profile bound to that instrument model. Profiles are versioned; when one is corrected, the stored raw can be re-parsed and the measurements regenerated.

---

**Rejected:** parsing at the edge of the wire. A parser bug then destroys data instead of merely mis-reading it, and every new instrument model becomes an agent release rather than a configuration entry.



**FIGURE 2 – Capture path.** The accented store is the one component that must never be edited or deleted. Everything downstream of it is reproducible; nothing upstream of it can be recovered. Adding a new lab, or a new instrument model, is a new parser profile rather than a code change – which is what makes the product resellable.

## Binding a reading to a sample

Capturing a weight is easy. Attaching it to the *right* sample is where these systems fail. Two supported strategies, both configuration-level choices:

- **Worklist-driven.** The operator selects or scans the subject, the screen shows what is expected, then presses print on the balance. The reading binds to what is on screen.
- **Sequence-driven.** The operator works positions one to fifty in order and the system assigns readings in sequence, with a confirmation step before commit.

In both cases the operator screen shows the expected subject and the live value together, and nothing commits without a confirmation. This single interaction rule prevents the most common and most expensive failure mode in an automated lab: fifty correct weights against forty-nine correct samples.

Version one pulls results only. Pushing worklists to instruments is worth doing eventually but roughly triples the integration surface per instrument, and none of the value described on the visit depends on it.

05

## Field capture and the connectivity problem

The concern raised was that there is no network where geologists work. The reframe that dissolves most of it: **the data does not need a network, it needs to arrive no later than the samples do.** The samples themselves travel physically, hours or days after collection. So the dispatch, not the individual sample, is the unit of synchronisation, and there are three independent channels.

#### DECISION D5

**The geologist's device syncs to the edge server over the lab's local network on arrival. A printed QR manifest is the fallback. Sample receipt is never blocked by missing field data.**

The primary channel involves no internet at all — the device and the server are on the same wireless network for the few minutes it takes. If the device does not arrive, the dispatch manifest travels on paper with a QR code carrying the payload, and is scanned at receipt.

If neither lands, receipt still proceeds and creates provisional sample records. When the dispatch arrives by any channel it reconciles against what was physically received and produces a discrepancy report: missing samples, unexpected samples, identifier mismatches.

---

**Rejected:** requiring the dispatch before samples can be logged in. It converts a connectivity inconvenience into a stopped laboratory, which is exactly the failure the client is trying to avoid.

That discrepancy report is worth calling out to the client as a feature in its own right. It is reconciliation they perform today by eye, against a paper sheet, under time pressure — and it is where sample mix-ups are currently caught, or not.

Geology is on paper today, which means the field application is a change-management problem as much as a technical one. Geologists are the users least subject to the lab manager's authority and the first who have to change habit. Plan the rollout so the lab gains value even if field adoption lags: the system must run correctly with dispatches keyed in at receipt, and get better when the field app arrives.

06

## Traceability and the audit trail

The requirement stated on the visit was that every interaction is logged — who, when, what action — and that a supervisor can see any sample's position in the chain at any moment. Both fall out of the model above, provided the log is built correctly from the start.

#### DECISION D6

**The audit log is append-only and hash-chained. Results are never updated in place; corrections create a superseding version with a reason.**

Each entry records actor, role, timestamp, action, entity, before and after values, terminal, and where required a reason code. Each entry includes a hash of the previous entry, so any tampering with history is detectable rather than merely discouraged.

Approval is an electronic signature: re-authenticate, capture the meaning of the signature, and bind it to the exact data version approved. A later correction does not silently invalidate an old approval — it supersedes it, visibly.

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**Rejected:** a distributed ledger or blockchain. Hash-chaining inside the database gives the tamper-evidence that is actually required, at a fraction of the operational cost. Reconsider only if an external party needs to verify records without trusting us.

A standard to stop citing

Earlier drafts of this project referenced **21 CFR Part 11**. That is US FDA pharmaceutical regulation and has no standing in a mining assay laboratory. Citing it in front of a chief chemist reads as a pharmaceutical template with the labels changed.

The frameworks that do apply: **ISO/IEC 17025** for laboratory competence, the **ALCOA+** data-integrity principles for what the audit trail must guarantee, **AMIRA P754** where metal accounting and mass balance enter scope, and the resource reporting codes — **JORC** or **NI 43-101** — which are what makes defensible assay data a corporate obligation rather than a lab preference.

The lab is not currently accredited, as far as we know. Design to 17025 anyway: it costs little at this stage, it is the reason the client wants a LIMS in the first place, and it is the single feature that makes the product sellable to a commercial laboratory later.

07

## Sample identity and labelling

There are no barcodes or scanners on site today. That is a recommendation to make, not a blocker — but the system must work fully without them, because they will not appear on day one.

The identifier scheme is configuration: prefix, site code, year, sequence, and a **check digit**. The check digit matters disproportionately. It costs nothing, and it means a mistyped or misread identifier fails immediately at the point of entry instead of silently attaching a result to the wrong sample.

#### DECISION D7

**Identity in the furnace is carried by tray and position, never by a label on the crucible.**

Paper and synthetic labels survive the drying oven at around 105 °C. Nothing survives a fusion furnace at 1000 °C. From the moment a charge is weighed into a crucible until the prill is weighed, the only identity is the tray identifier plus the position number — which is a further reason positions are first-class objects rather than list indices.

### What to recommend on hardware

- **Printers.** Thermal-transfer with resin ribbon on synthetic label stock for the prep area. Direct-thermal labels fade in heat and are the usual reason a lab concludes that barcodes do not work.
- **Scanners.** Two-dimensional imagers in USB keyboard-wedge mode. They need no drivers and no integration work — the scan simply arrives as keystrokes in the focused field, which means scanning can be added to the system after go-live without changing a line of code.
- **Symbology.** Code 128 for linear labels on bags and pulp packets; QR where the label is small or must carry more than an identifier, including the dispatch manifest fallback in section 5.

Sequence the introduction so the lab is not asked to change two things at once: version one prints labels and accepts typed identifiers; scanning is enabled per station as hardware arrives.

#### 08

### Doré and bullion, through to purity

Scope extends past the assay bench to the purity of the poured bar — **doré and refined bullion alike** — which adds a second, shorter chain: a **melt** produces one or more **bars**, each with a serial and gross mass; each bar is sampled — typically by drillings or a dip sample — and that sample re-enters the assay chain to produce a **fineness**.

Two boundaries are worth being precise about, because earlier material on this project got them wrong and they are the kind of error that costs credibility in the room.

- Doré and bullion are modelled **identically** — a melt, a bar, a bar sample, a fineness. What differs is who certifies the bar for sale. London Good Delivery is a *refiner* accreditation, and serials on Good Delivery bars are assigned by the accredited refiner. Where an operation refines and sells its own bars the system holds the assay evidence behind the certificate; it does not confer the accreditation, and should never be pitched as though it does.

- The valuable capability at this end of the chain is **outturn reconciliation** — comparing the mine's own doré assay against the refiner's return assay, tracking the variance, and holding the evidence when the two disagree. That difference is money, it is currently reconciled in spreadsheets, and no incumbent product sells it well.

Bars link back to their source melts, and melts to the production period they came from, so that the mass balance in section 12's later phases has something to close against.

09

## Reports and outputs

The lab issues Excel workbooks and branded PDFs on company letterhead. Both come from one dataset through a template engine, with branding as tenant configuration — a second lab supplies its own letterhead rather than receiving a code change.

Three rules that are cheap now and awkward later:

- A report is a **versioned issue**. A re-issued certificate supersedes its predecessor visibly, carrying the reason. It never silently replaces it.
- Re-assayed results carry their flag **into the report**, as the client requested, not merely in the database.
- Machine-readable exports are first-class alongside human-readable ones, because the geology database is the real consumer of this data — see the next section.

10

## Integration edges

### acQuire, upstream and downstream

Geology mentioned acQuire; the lab was working in Excel. This is the most important integration boundary in the project, and the one we know least about. acQuire is where drill hole and sample data legitimately lives, and where assay results have to land for resource estimation to use them. Our system should be the authority on what happened *in the lab*, and acQuire the authority on what happened *in the ground* — with a defined exchange in both directions rather than a duplicated master.

Practically: dispatches can originate from acQuire as expected-sample lists, and results return as import files in an agreed layout. We need the acQuire product and version, and whether geology uses its laboratory workflow, before committing to a design here. Treat it as a spike, not an assumption.

## The end-of-day server

The client referred to a server where everything is dumped at the end of the day. Nobody yet knows what it is. It could be a file share, a backup target, or an existing system with a database worth reading. Finding out is a half-day of work and could change the migration plan materially.

## Migration

Historical data arrives as Excel and CSV. Build an importer with column mapping, validation, a dry run, and a reconciliation report — not a one-off script.

### DECISION D8

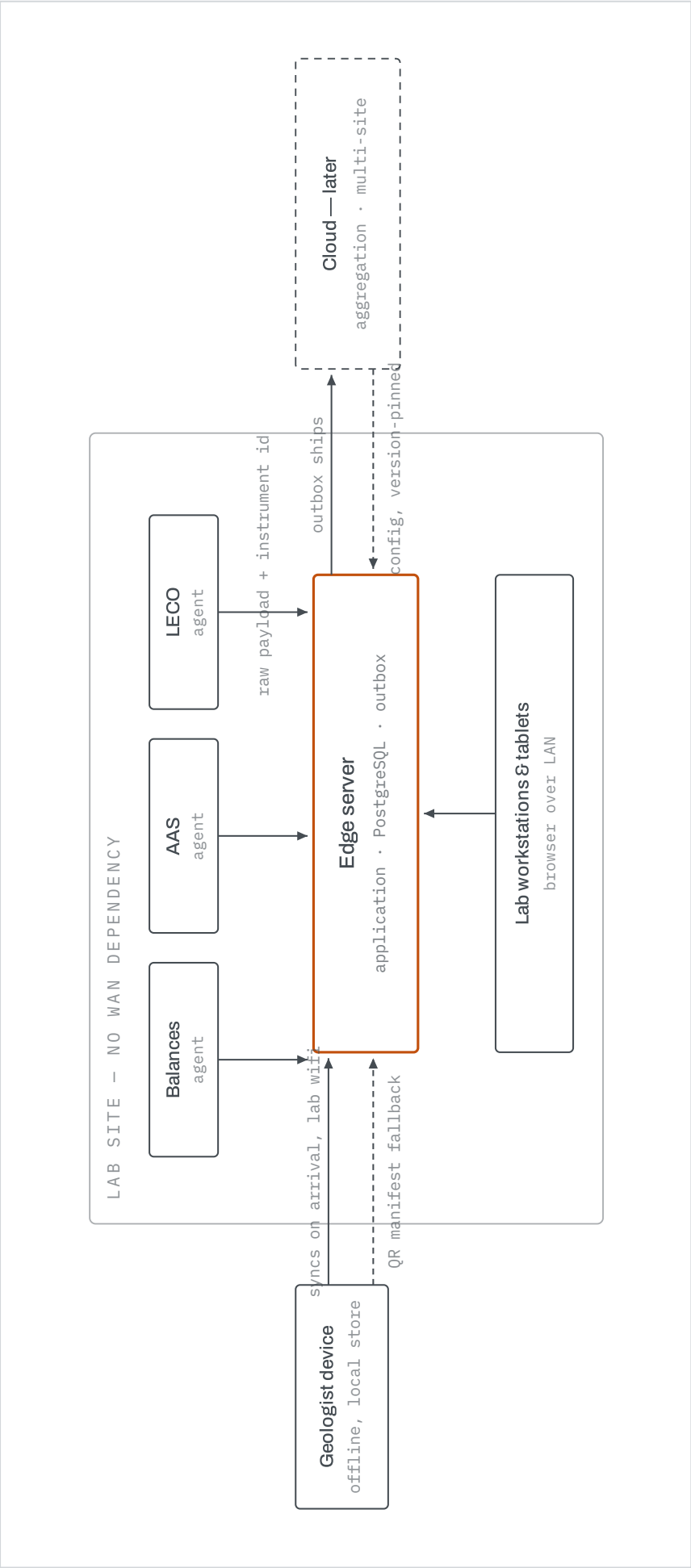
**Migrated records are marked as legacy, carry their source file as provenance, and are read-only.**

Imported history has no instrument capture behind it, no audit trail, and no QC evaluation. If it is stored indistinguishably from data the system produced, every integrity guarantee in section 6 becomes unprovable, because a reader cannot tell which records were captured and which were typed.

11

## Topology and stack

The client's position on cloud hosting is not yet known, and mining IT policy — not our preference — will decide it. So the system is built to run completely without one, with the cloud as an addition rather than a dependency.



**FIGURE 3 – Deployment.** Everything inside the site boundary works with the wide-area link down, indefinitely. The two dashed paths are the ones that tolerate absence: the manifest fallback when a device does not arrive, and configuration flowing down from a cloud tier that may never be built.

#### DECISION D9

**Synchronisation is resolved by ownership, not by merge. The site owns lab records; the cloud owns configuration.**

Lab data flows upward from an append-only outbox and is read-only above. Configuration flows downward, version-pinned, and is applied explicitly at the site. Two directions, two owners, and consequently almost no true conflicts to resolve.

---

**Rejected:** conflict-free replicated data types, proposed in earlier drafts. CRDTs solve concurrent editing of shared state. Lab records have a single writer by definition – one operator, one instrument, one moment – so we would be paying a large complexity cost to solve a problem the domain does not have.

---

**Rejected:** PostgreSQL and CouchDB side by side. Two stores means two truths and a reconciliation layer nobody owns. One relational store with an outbox does the job.

#### DECISION D10

**A modular monolith on PostgreSQL, not microservices.**

One deployable application with firm internal module boundaries – capture, core chain, QC, reporting, sync. Ten to fifteen users at one site do not need a distributed system, and a distributed system is considerably harder to install on a server in a mine lab and support remotely. The module seams are where services can be split later, if scale ever justifies it.

### Stack

- **Data.** PostgreSQL as the single source of truth. Relational integrity for the chain, JSONB for configuration objects and raw payloads.
- **Capture agent.** A single self-contained binary running as a Windows service. Go or .NET, chosen for the serial-port story and for the fact that it must install on whatever Windows version the instrument PCs turn out to be running.
- **Application.** One backend framework the team already knows. This is not the place to learn something – the domain is the hard part.
- **Lab client.** A browser application, built for touch and gloved hands: large targets, high contrast, no hover-dependent controls.
- **Field client.** A mobile application with a local database and a sync client, built offline-first from the start rather than retrofitted.

Multi-tenancy should be present in the schema from the first migration – a tenant identifier on every row – even while only one tenant exists. Adding it later means rewriting every query in the system.

## Scope by release

Sequenced by dependency and by what the client will judge the system on. Version one has to make the lab visibly better at the two things they raised first: weights arriving without being typed, and knowing where a sample is.

| CAPABILITY                                             | RELEASE | WHY THERE                                                                       |
|--------------------------------------------------------|---------|---------------------------------------------------------------------------------|
| Sample receipt, reconciliation, status tracking        | V1      | The visible win. Answers “where is this sample” on day one.                     |
| Transformation chain with mass in and out              | V1      | The core model. Everything else attaches to it.                                 |
| Tray batching with configurable QC insert map          | V1      | Positions are unrecoverable if not captured at the time.                        |
| Balance capture via agent                              | V1      | Highest value per unit of effort. The client asked for this first.              |
| AAS and LECO file capture, parser profiles             | V1      | Completes the no-typing promise.                                                |
| Configurable QC rules, control charts, hold on failure | V1      | The reason a lab trusts its own numbers.                                        |
| Blind control visibility and the two QC streams        | V1      | Touches every screen and query that shows a sample. Cannot be added afterwards. |
| Approval, segregation of duties, re-assay flagging     | V1      | Explicitly requested. Cheap now, structural later.                              |
| Hash-chained audit trail                               | V1      | Cannot be retrofitted over data already written.                                |
| ID scheme with check digit, label printing             | V1      | Scanners can follow; identifiers cannot be restated.                            |
| Field dispatch app and local sync                      | V1      | Removes the paper re-keying that starts every error.                            |
| Excel and branded PDF reporting                        | V1      | Matches what the lab issues today. Adoption depends on it.                      |
| Doré and bullion — melt, bar, bar sample, fineness     | V1      | In scope per the client. The chain ends at a purity on a numbered bar.          |
|                                                        |         |                                                                                 |

| CAPABILITY                                                    | RELEASE | WHY THERE                                                              |
|---------------------------------------------------------------|---------|------------------------------------------------------------------------|
| Excel and CSV migration importer                              | V1      | History has to come across for the system to be believed.              |
| acquire two-way exchange                                      | V2      | Needs discovery first. Interim: file export in an agreed layout.       |
| Consumable lots — crucibles, cupels, flux; firing counts      | V2      | Deferred by the client. CRM lots remain in V1.                         |
| Instrument and balance asset register with calibration status | V2      | Required for accreditation, not for daily operation.                   |
| Refiner outturn reconciliation                                | V2      | High commercial value; needs real outturn documents to design against. |
| External and umpire lab dispatch and comparison               | V2      | Unknown whether the client does this. Question 17 outstanding.         |
| Cloud aggregation, multi-site, TAT dashboards                 | V2      | One site today; schema is ready for it from V1.                        |
| Commercial lab module — clients, jobs, invoicing              | V3      | The resale path. Not needed by an in-house lab.                        |
| Metal accounting and mass balance                             | V3      | Crosses into plant systems. A separate conversation.                   |
| Worklist push to instruments                                  | V3      | Roughly triples integration surface per instrument.                    |
| Anomaly detection on assay and mass patterns                  | V3      | Needs a year of the system's own clean data before it means anything.  |

13

## Risks

| RISK                                                      | LEVEL | RESPONSE                                                                                                                                                                                  |
|-----------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Instrument makes, models and output formats still unknown | HIGH  | The largest schedule variable in the project. One timeboxed spike per instrument before any estimate is given. Until then, integration effort is unquantified.                            |
| Instrument PCs running old or locked-down Windows         | HIGH  | Common project-killer. Confirm the OS version and installation rights on each PC before committing to the agent design; keep the watched-folder mode available as the low-privilege path. |

| RISK                                                                                    | LEVEL | RESPONSE                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| acquire exchange format and version unknown                                             | HIGH  | Discovery session with the geology team. Ship a file-based export in V1 so the project is not blocked on it.                                                                                                            |
| Unclear whether the operation pours refined bullion on site or doré for onward refining | MED   | Changes nothing in the data model — both are melt, bar, sample, fineness — but it decides whether outturn reconciliation against a third-party refiner is a V2 requirement or irrelevant. One question to the met team. |
| Throughput unknown — samples per shift, batches per day                                 | MED   | Affects sizing and UI density, not the architecture. Obtain before hardware is specified.                                                                                                                               |
| Geologists moving from paper to a device                                                | MED   | Sequence so the lab benefits first. Design receipt to work with keyed-in dispatches.                                                                                                                                    |
| Site power and network reliability, backup regime                                       | MED   | Edge server needs a UPS and a tested restore. Unknown today; ask during the IT session.                                                                                                                                 |
| Quality of historical Excel data                                                        | MED   | Dry-run import with a reconciliation report before anything is committed. Legacy records stay read-only.                                                                                                                |
| Cloud posture undecided by client IT                                                    | LOW   | Already mitigated by D9 — the system is complete without a cloud tier.                                                                                                                                                  |
| No barcode hardware on site                                                             | LOW   | Keyboard-wedge scanners need no integration. Typed identifiers with a check digit work in the meantime.                                                                                                                 |

A

## What we need from the geology team

Written to be forwarded as-is. It avoids software vocabulary, because the people answering are geologists, not system users yet.

*We are building the laboratory system that will receive your samples. Before we design how field data reaches the lab, we need to understand how you work now. Nothing here is a criticism of the current process — we simply need to fit around it rather than against it.*

### 01 What do you use to record hole and sample details?

acquire, Excel, a paper log book, or something else? If acquire, which version, and do you use its laboratory or dispatch workflow?

### 02 Can we see a completed submission sheet?

A photograph of a real one that went to the lab recently, filled in, is more useful to us than a blank template.

**03 What do you record for each sample?**

Hole identifier, from and to depth, sample number, sample type, date, who collected it — and anything else that has to reach the lab with the samples.

**04 Tell us about the control samples you put into a dispatch.**

Blanks, certified standards or field duplicates that you add before samples go to the lab. How many per hundred, which standards and whose certificates, and should the lab be able to tell which ones they are? The system will support hiding them from the lab — we need to know whether that should be the default, and who on your side reviews the results when a control fails.

**05 How does the paperwork physically reach the lab today?**

Carried with the samples, emailed ahead, or typed in by lab staff on arrival?

**06 How do you want results back?**

What format, and does it need to load straight into your database without anyone re-typing it? A copy of a results file you have loaded successfully before would help.

**07 What do you actually have with you in the field?**

Phones, tablets, laptops. Where do you get signal or wifi — at the drill site, back at camp, only at the office?

**08 How many samples in a typical dispatch, and how many dispatches a week?**

Rough numbers are fine. We need the busiest week, not the average one.

**09 Who assigns sample numbers, you or the lab?**

And what is the format? If there is an existing numbering convention we must not break, tell us now.

**10 What happens when the samples do not match the sheet?**

A sample is missing, an extra one turns up, or a number is unreadable. Who notices, and what do they do about it?

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Draft for internal review. Decisions D1-D11 are the parts that are expensive to change once code exists; everything else is negotiable. Outstanding inputs are listed in section 13 and Appendix A — instrument models and the acQuire configuration are the two that gate estimation.